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College of Informatics and Computing Sciences

PROJECT PROPOSAL
OOP 101 – OBJECT-ORIENTED PROGRAMMING
Computer Science Department
1st Semester, AY 2026–2027

I. TEAM INFORMATION

Team Name: Lost And Found			
Team Number:6		Section:BSIT-2108	
No.	**SR-Code**	**Name**	**GitHub Username**
1	25-00347	Garing, Kristine Joy I.	25-00347-cmyk
2	25-09411	Panganiban, Kerk Daniel M.	25-09411-netizen
3	25-00016	Panganiban, Jhanel C.	25-00016-max
4	25-02835	Viceral, Jhayzel Liann B.	jhayzel

List the names in alphabetical order, last name first. If there is no 4th member, remove the row.

II. PROJECT INFORMATION

Project Title

BATSTATEU CAMPUS LOST AND FOUND ITEM TRACKING AND MANAGEMENT SYSTEM

Project Background

Losing personal belongings inside the school campus can be a problem for students, teachers, and other school staff. Sometimes, people find lost items but do not know who owns them or where to report them. It can also be difficult to find missing belongings when there is no organized record of lost and found items.

The BATSTATEU Campus Lost and Found Item Tracking and Management System aims to make this process easier by providing a simple Java-based application where users can report lost and found items, view available records, and search for possible matches.

This system will help organize item information and make it easier to manage lost and found records within the Batangas State University campus.

Guide Questions:

- What problem are you addressing?
- Who experiences the problem?
- Why is it important to solve?

Project Objectives

General Objective

To develop a Java-based application that will help organize and manage lost and found items within the Batangas State University campus.

Specific Objectives
The project aims to:
1. Allow users to report lost and found items by entering the required information.
2. Display all recorded lost and found items.
3. Provide a search feature to help users find specific items.
4. Show possible matches between reported lost and found items.
5. Allow the management of item records, including updating their status and removing records.
6. Apply Java programming and Object-Oriented Programming concepts in developing the system.

Target Users
Identify who will use the application.

Target User	How They Will Use the System
Students	Can report lost belongings, report found items, and search for missing items.
Faculty and School Staff	Can report found items and help manage lost and found records within the campus.

Proposed Features
List the main features of the application.

Feature	Description
Report Lost Item	Allows users to enter information about an item they have lost.
Report Found Item	Allows users to record information about an item they have found.
View All Records	Displays the list of recorded lost and found items.
Search and Find Possible Matches	Allows users to search for items and view possible matches between lost and found records.
Manage Item Records	Allows the system to update item status, process claim requests for verification, and remove records.

III. JAVA AND OOP CONCEPTS
Explain how the project will use the concepts learned in OOP 101.

OOP Concept	How It Will Be Used
Classes and Objects	The system uses classes such as Item, LostItem,FoundItem, LostAndFoundSystem, and Main. Objects are created from these classes to represent items and handle system operations.
Methods	Methods are used to perform different tasks, such as adding items, displaying records, searching for items, updating status, and removing records.
Encapsulation	The item information is kept private inside the Item class. Getters and setters are used to access or update the information when needed.

Inheritance	LostItem and FoundItem inherit common information and behavior from the parent class Item. This avoids repeating the same code in both classes.
Polymorphism	The displayInfo() method is overridden in the LostItem and FoundItem classes. This allows each type of item to display its information based on its own class.
Abstraction	The Item class is an abstract class. It provides the common structure for lost and found items without being used directly to create an ordinary item object.
Exception Handling	The system uses input validation and exception handling to prevent invalid user input from interrupting the program.
Arrays/Data Handling	An ArrayList is used to temporarily store and manage lost and found item objects. It allows the system to add, display, search, update, and remove records.

Note: The team should use OOP concepts where they are appropriate to the project. Do not add features just to force a concept into the system.

IV. PROPOSED CLASSES

List the main classes that the project may need.

Class	Purpose	Main Attributes	Main Methods
Item	Abstract Parent class that contains the common information of lost and found items.	Private item information and status	Getters, setters, displaceInfo()
Lost Item	Represents an item that has been reported lost. it inherits from Item.	Inherited item information	Overridden displayInfo()
Found Item	Represents an item that has been reported found. It inherits from Item.	Inherited item information	Overridden displayInfo()
LostAndFoundSystem	Handles the main operations of the application and manages item records.	ArrayList<Item>	Add item, display records, search items, find possible matches, update status, remove record
Main	The starting point of the application. It displays the menu and accepts user inputs.	Scanner and system object	main() and menu/input handling

The classes may be changed as the project is developed.

V. PROGRAM FLOW

Briefly describe how the user will use the application.

Start
↓
Display System Title
↓
Display Main Menu
↓
User Selects an Operation
↓

- Report Lost Item
- Report Found Item
- View All Records
- Search Item / Find Possible Matches
- Update Item Status
- Remove Item Record
- Exit

↓
If the user selects an operation, enter the required information.
↓
Validate User Input
↓
Process the Selected Operation
↓
Display the Result or Confirmation Message
↓
Return to Main Menu
↓
If the user selects Exit
↓
End Program

VI. TEAM RESPONSIBILITIES

Member	Assigned Tasks
Team Leader	Coordinates Task, Monitors deadlines, Communicates with instructor.
Lead Programmer	Coordinates the Implementation of Java Classes, methods, and OOP Features
System Design Lead	Develops System Requirements, Class Design, UML Diagrams and overall application structure.
Documentation Lead	Maintains the README, Testing Documentation, Screenshots and assist in checking the system functionality.

VII. DEVELOPMENT SCHEDULE

Week	Activity	Expected Output
Week 1	Planning and requirements	Project proposal
Week 2	System and class design	Class/UML diagram

Week 3	Initial coding	Basic classes
Week 4–5	Feature development	Working features
Week 6	OOP implementation	Completed OOP features
Week 7	Testing and debugging	Tested application
Week 8	Documentation and presentation	Final project

The schedule may be adjusted based on the course calendar.

VIII. EXPECTED OUTPUT

Briefly describe what the completed application should do. You can include screenshots.
The team expects to produce a functional Java application that [describe the main purpose of the system].
The application will demonstrate the appropriate use of Java and object-oriented programming concepts.

IX. PROJECT LIMITATIONS

List features that are not included in the project.

X. GITHUB REPOSITORY

The repository will contain the source code, README, documentation, UML/class diagram, screenshots, and other required project files.

Repository Name: BatStateU-Campus-Lost-and-Found-Item-Tracking-Management-System

Link to Repository:
https://github.com/25-09411-netizen/BatStateU-Campus-Lost-and-Found-Item-Tracking-Management-System

XI. APPROVAL

Prepared by:

Name	Signature